# Recent updates to `nebulizer`: command-line utility for remote Galaxy admin tasks

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## Introduction

**Nebulizer is a Python utility which provides a high-level interactive command line interface to remotely administer Galaxy servers[1].**

- Enables admin operations to be performed efficiently via the command line, as an alternative to using a Galaxy instance's web interface
- Developed to help "part-time" admins perform day-to-day administrative tasks across multiple Galaxy instances
- Uses Bioblend[2] to interact with instances via their APIs & complements lower-level interfaces of Ephemeris[3] and Parsec[4]

## Installation & usage

Easy to get started: for example, to install from PyPI into Python virtualenv:

```
virtualenv venv
source venv/bin/activate
pip install nebulizer
```

Provides a single command 'nebulizer' with subcommands for different operations e.g.

```
nebulizer ping https://usegalaxy.org/
```

Easily work with multiple Galaxy instances by storing API keys against aliases e.g.

```
nebulizer add_key galaxy main \
     https://usegalaxy.org/
```

**Nebulizer Galaxy admin commands**

## User management

- `create_user`: adds an individual Galaxy user
- `create_batch_users`: adds users in a batch
- `list_users`: gets information about users in Galaxy including status, disk and quota usage
- `delete_user`: remove a user account

## Tool management

- `list_tools`: get information about tools including versions and status
- `[install | update | uninstall]_tool`: manage installed tools in a Galaxy instance
- `search_toolshed`: find tools to install

## Quota management

- `quotas`: lists quotas in a Galaxy instance
- `quota_add | quota_mod | quota_del`: create, modify and delete quotas

## Data library management

- `create_library[_folder] & add_library_dataset`: create & populate data libraries and folders
- `list_libraries`: query data library contents

## General purpose functions

- `ping`: tests availability of Galaxy instance
- `config`: queries Galaxy configuration

## Recent developments

- New quota management functionality
- ⚠ Substantial refactoring and simplification of tool management commands (only `list_tools`, `install_tool`, `update_tool` and `uninstall_tool` now supported)
- Tool update accepts wildcards (to update multiple tools with one command)
- Filter users on status & sort on disk/quota usage
- ⚠ Dropped support for Python 2.7

## Availability

- **Docs** https://nebulizer.readthedocs.io
- **Github** https://github.com/pjbriggs/nebulizer
- **PyPI** https://pypi.org/project/nebulizer/
- **Bioconda** https://anaconda.org/bioconda/nebulizer

## Acknowledgements

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Nebulizer ported to Bioconda by Peter van Heusden.

## References

**[1] The Galaxy platform for accessible, reproducible and collaborative biomedical analyses: 2018 update.** Afgan *et al., Nucleic Acids Research*, 2018, 46(W1), W537–W544; doi:10.1093/nar/gky379

**[2] BioBlend: automating pipeline analyses within Galaxy and CloudMan**. Sloggett *et al., Bioinformatics*, 2013, 29(13), 1685-1686; doi:10.1093/bioinformatics/btt199

**[3] Ephemeris** https://ephemeris.readthedocs.io/

**[4] Parsec** https://parsec.readthedocs.io/

**Run through the Nebulizer tutorial at https://nebulizer.readthedocs.io/en/latest/tutorial.html**